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15. IMPLEMENT IRIS FLOWER CLASSIFICATION USING NAIVE BAYES CLASSIFIER

1. Iris Dataset 1

Dataset: iris1.csv

Sepal Length	Sepal Width	Petal Length	Petal Width	Species
5.1	3.5	1.4	0.2	Setosa
4.9	3.0	1.4	0.2	Setosa
6.0	2.9	4.5	1.5	Versicolor
5.9	3.0	4.2	1.5	Versicolor
6.5	3.0	5.2	2.0	Virginica
6.7	3.1	5.6	2.4	Virginica

PROGRAM

```
import pandas as pd

from sklearn.naive_bayes import GaussianNB

df = pd.read_csv("iris1.csv")

X = df.iloc[:, :-1]

y = df.iloc[:, -1]

model = GaussianNB()

model.fit(X, y)

sample = [[6.4, 3.0, 5.1, 1.9]]

print("Predicted Flower:", model.predict(sample)[0])
```

OUTPUT

Predicted Flower: Virginica

2. Iris Dataset 2

Dataset: iris2.csv

Sepal Length	Sepal Width	Petal Length	Petal Width	Species
5.0	3.4	1.5	0.2	Setosa
5.2	3.5	1.5	0.2	Setosa
5.8	2.7	4.1	1.0	Versicolor
6.1	2.8	4.0	1.3	Versicolor
6.3	2.9	5.6	1.8	Virginica
6.7	3.0	5.2	2.3	Virginica

PROGRAM

```
import pandas as pd
from sklearn.naive_bayes import GaussianNB
df = pd.read_csv("iris2.csv")
X = df.iloc[:, :-1]
y = df.iloc[:, -1]
model = GaussianNB()
model.fit(X, y)

sample = [[5.1, 3.4, 1.5, 0.2]]
print("Predicted Flower:", model.predict(sample)[0])
```

OUTPUT

Predicted Flower: Setosa

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3. Iris Dataset 3

Dataset: iris3.csv

Sepal Length	Sepal Width	Petal Length	Petal Width	Species

5.4	3.7	1.5	0.2	Setosa
4.8	3.4	1.6	0.2	Setosa
6.0	2.7	5.1	1.6	Versicolor
5.7	2.8	4.5	1.3	Versicolor
6.2	3.4	5.4	2.3	Virginica
6.4	3.2	5.3	2.3	Virginica

PROGRAM

```
import pandas as pd
from sklearn.naive_bayes import GaussianNB
df = pd.read_csv("iris3.csv")
X = df.iloc[:, :-1]
y = df.iloc[:, -1]
model = GaussianNB()
model.fit(X, y)
sample = [[5.9, 2.8, 4.7, 1.2]]
print("Predicted Flower:", model.predict(sample)[0])
```

OUTPUT

Predicted Flower: Versicolor

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4. Iris Dataset 4

Dataset: iris4.csv

Sepal Length	Sepal Width	Petal Length	Petal Width	Species
5.1	3.8	1.5	0.3	Setosa
4.7	3.2	1.3	0.2	Setosa
6.2	2.9	4.3	1.3	Versicolor
5.6	2.9	3.6	1.3	Versicolor

6.8	3.2	5.9	2.3	Virginica
7.1	3.0	5.9	2.1	Virginica

PROGRAM

```
import pandas as pd
from sklearn.naive_bayes import GaussianNB

df = pd.read_csv("iris4.csv")
X = df.iloc[:, :-1]
y = df.iloc[:, -1]
model = GaussianNB()
model.fit(X, y)
sample = [[6.9, 3.1, 5.8, 2.2]]
print("Predicted Flower:", model.predict(sample)[0])
```

OUTPUT

Predicted Flower: Virginica

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5. Iris Dataset 5

Dataset: iris5.csv

Sepal Length	**Sepal Width**	**Petal Length**	**Petal Width**	**Species**
5.0	3.5	1.3	0.3	Setosa
5.3	3.7	1.5	0.2	Setosa
5.9	3.0	4.2	1.5	Versicolor
6.0	2.9	4.5	1.5	Versicolor
6.5	3.2	5.1	2.0	Virginica

6.7	3.3	5.7	2.1	Virginica

PROGRAM

```
import pandas as pd
from sklearn.naive_bayes import GaussianNB
df = pd.read_csv("iris5.csv")
X = df.iloc[:, :-1]
y = df.iloc[:, -1]

model = GaussianNB()
model.fit(X, y)
sample = [[5.8, 3.0, 4.1, 1.4]]
print("Predicted Flower:", model.predict(sample)[0])
```

OUTPUT

Predicted Flower: Versicolor